

A Rectangle Counterexample to Weighted Log-Submodularity

Automated mathematics run `fa_banach_001`, lane 16

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Status. Candidate counterexample, likely valid, pending expert review. The conjecture following equation (44) of Fradelizi–Langharst–Madiman–Zvavitch is false for a full-dimensional log-concave probability measure on $\mathbb{R}^2$ and three full-dimensional symmetric zonotopes.

1 Source conjecture

The current version of [1] asks for which measures μ and classes of convex sets one has

$$\mu(A)\mu(A + B + C) \leq \mu(A + B)\mu(A + C), \quad (1)$$

and conjectures that (1) holds whenever A, B, C are symmetric zonoids and μ is any log-concave measure. This appears on page 27, immediately after equation (44).

Theorem 3.25. *(The case of log-concave measures in $\mathbb{R}$) Let μ be a log-concave measure on $\mathbb{R}$. Then, for any interval $A \subset \mathbb{R}$ and any symmetric intervals $B, C \subset \mathbb{R}$, one has*

$$\mu(A)\mu(A + B + C) \leq \mu(A + B)\mu(A + C).$$

In the volume case, it is known that (42) with $c_{\mu,n} = 1$ for $n \geq 3$ is false. We leave determining which measures and which classes of convex sets satisfy (42) with $c_{\mu,n} = 1$ as an open question. That is, if $\mathcal{C}$ is some fixed class of convex sets, we ask for which measures μ does it hold that, for $A, B, C \in \mathcal{C}$, one has

$$\mu(A)\mu(A + B + C) \leq \mu(A + B)\mu(A + C). \quad (44)$$

We conjecture that (44) holds for A, B, C being symmetric zonoids and μ being any log-concave measure. In the volume case, log-submodularity with constant 1 was established recently by a subset of the authors, working with Meyer, in $\mathbb{R}^2$ and $\mathbb{R}^3$ [18] when A, B and C are taken to be zonoids; the 2-dimensional case follows from [43].

We conclude this subsection by showing that (44) is related to the so-called Bézout-type inequalities for mixed measures (see [16]).

Lemma 3.26 (Local form of log-submodularity). *Fix $n > 1$. Let μ be a Radon measure on*

Figure 1: The open question and conjecture in [1, p. 27].

2 Proof intuition

The conjecture allows the measure to be highly anisotropic. Concentrate a log-concave measure in a thin parallelogram around the diagonal $x = y$. Horizontal enlargement alone and vertical enlargement alone

barely increase the amount of diagonal mass captured by a centered square. Applying both enlargements captures a much longer diagonal interval. Centered rectangles are already zonotopes, and a width-1/10 parallelogram makes this mechanism an exact full-dimensional area computation rather than a limiting argument.

3 Counterexample

Theorem 1. *The conjecture following equation (44) of [1] is false as stated. The failure occurs in dimension two with an absolutely continuous log-concave probability measure and full-dimensional symmetric zonotopes.*

Proof. Set $\delta = 1/10$ and consider the invertible linear map

$$T(u, v) = (u + v, u - v).$$

Let

$$P = T([-4, 4] \times [-\delta, \delta])$$

and let μ be normalized two-dimensional Lebesgue measure on P :

$$\mu(E) = \frac{\text{area}(E \cap P)}{\text{area}(P)}.$$

The density of μ is a positive constant times the indicator of the convex body P , hence μ is a log-concave probability measure. Equivalently, this follows directly from the Brunn–Minkowski inequality applied to $P \cap E$ and $P \cap F$.

Define the centered rectangles

$$\begin{aligned} A &= [-1, 1] \times [-1, 1], \\ B &= [-1, 1] \times [-\delta, \delta], \\ C &= [-\delta, \delta] \times [-1, 1]. \end{aligned}$$

They have nonempty interior and are origin-symmetric zonotopes, hence symmetric zonoids.

We compute their measures and those of their Minkowski sums. Write $R(s, t) = [-s, s] \times [-t, t]$. In (u, v) coordinates, membership of $T(u, v)$ in $R(s, t)$ is equivalent to

$$|u + v| \leq s, \quad |u - v| \leq t. \tag{2}$$

The Jacobian magnitude of T is 2, and it cancels from all normalized-area ratios. The support rectangle in (u, v) coordinates has area 16δ .

For $R(q, q)$, the two inequalities in (2) are equivalent to $|u| + |v| \leq q$. For the values of q used below, its preimage inside the support has area

$$\int_{-\delta}^{\delta} 2(q - |v|) dv = 4q\delta - 2\delta^2.$$

Consequently,

$$\mu(R(q, q)) = \frac{4q\delta - 2\delta^2}{16\delta} = \frac{q}{4} - \frac{\delta}{8}. \tag{3}$$

Next, for $R(2, 11/10)$ the constraint $|u + v| \leq 2$ is redundant: indeed, $|u - v| \leq 11/10$ and $|v| \leq 1/10$ imply $|u + v| \leq 13/10$. Thus its preimage has area $2\delta \cdot 2(11/10)$, giving

$$\mu(R(2, 11/10)) = \frac{11}{40}. \tag{4}$$

The same value holds for $R(11/10, 2)$ by symmetry.

The relevant Minkowski sums are

D	D as a centered rectangle	$\mu(D)$
A	$R(1, 1)$	$19/80$
$A + B$	$R(2, 11/10)$	$11/40$
$A + C$	$R(11/10, 2)$	$11/40$
$A + B + C$	$R(21/10, 21/10)$	$41/80$

Here the square entries follow from (3) and the other two from (4). Therefore

$$\begin{aligned} & \mu(A)\mu(A + B + C) - \mu(A + B)\mu(A + C) \\ &= \frac{19}{80} \frac{41}{80} - \left(\frac{11}{40}\right)^2 = \frac{779 - 484}{6400} = \frac{59}{1280} > 0. \end{aligned}$$

This is the strict reverse of (1), and proves the claim. $\square$

4 Robustness and limitations

The measure is already full-dimensional and absolutely continuous. If a smooth, everywhere-positive density is desired, convolve μ with a centered nondegenerate Gaussian of variance ε^2 . Convolution preserves log-concavity [2]. As $\varepsilon \downarrow 0$, the four probabilities in the proof converge to their displayed values because rectangle boundaries have μ -measure zero. The defect is strictly positive, so it remains positive for all sufficiently small ε .

The argument refutes the quantifier “any log-concave measure.” It makes no claim about variants restricted to radial measures, the standard Gaussian, or Lebesgue measure. Those narrower questions may remain open.

5 Verification and novelty

The exact-rational script `code/check_exact_measures.py` reproduces the four entries of the table and the defect $59/1280$. It is only a transcription check; the proof is the preceding area calculation.

The local run indexes were searched for arXiv:2402.10314, the exact title, log-submodularity, symmetric zonoids, and log-concave measures. Bounded web searches on August 11, 2026 used the exact title with “counterexample,” the arXiv id with “log-submodularity conjecture,” and the authors with the core terms. They found the current v3 source and adjacent results for Lebesgue volume, but no later answer or this construction. Novelty confidence is positive but conservative pending expert review.

Human review should verify the four area ratios and confirm the literal scope of “any log-concave measure.” No lower-dimensional convention is involved: P, A, B, C all have nonempty interior in $\mathbb{R}^2$.

References

- [1] M. Fradelizi, D. Langharst, M. Madiman, and A. Zvavitch, *Weighted Brunn–Minkowski Theory II: Inequalities for Mixed Measures and Applications*, arXiv:2402.10314v3, 2026.
- [2] R. J. Gardner, *The Brunn–Minkowski inequality*, Bulletin of the American Mathematical Society **39** (2002), 355–405.